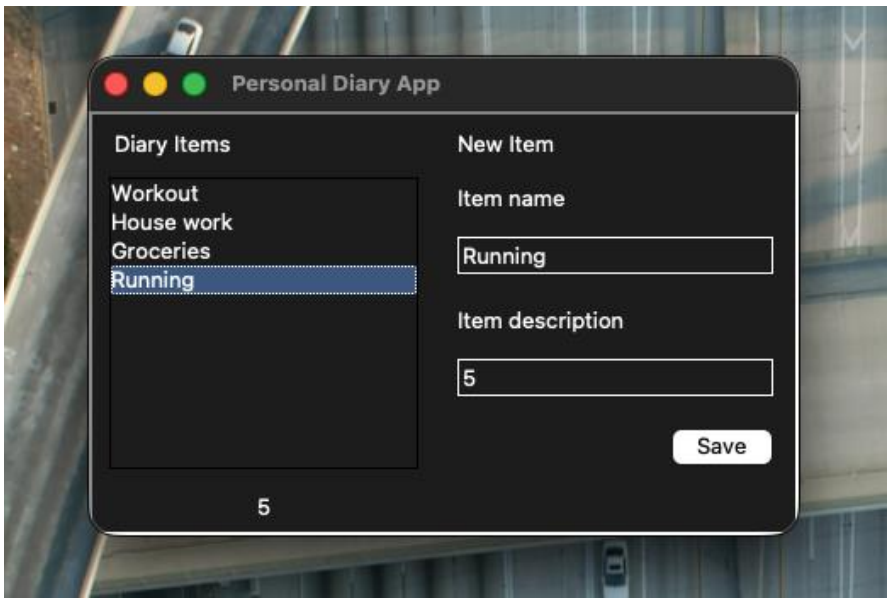
Assignment 2 Case Studies with Programming Problems

Case Study 1: Python Tkinter and Pillow packages

Explanation: In this task I learned how to build a simple GUI using Tkinter widgets and run sample applications like personal_diary_app.py.

Task 1: Introduction to Tkinter

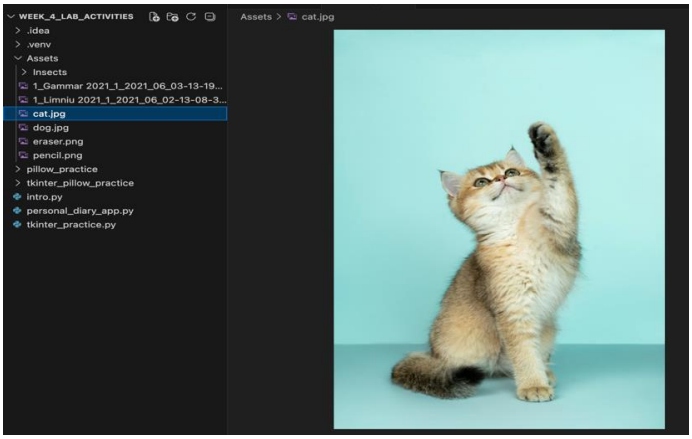
- I ran python3 personal_diary_app.py and then received this result.



Task 2:Introduction to Pillow library/package

Explanation: Install and use the Pillow library to load images, extract basic information (size, format), and perform simple operations like rotate and flip.

- Installation and setup for tasks



- Display the image of the cat

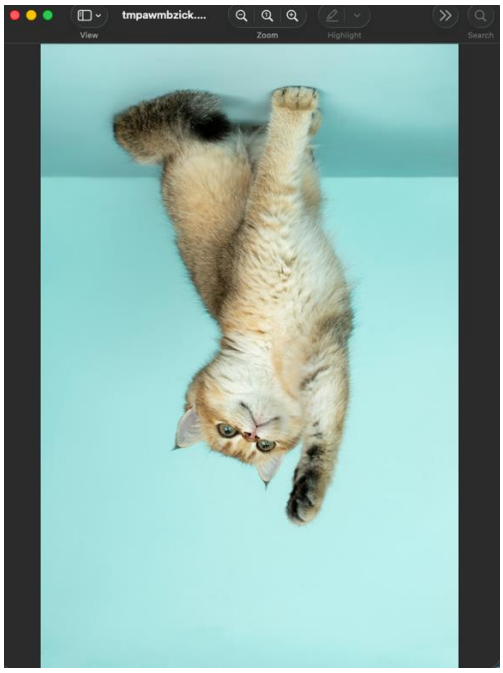
- Getting Basic Image Information

```
from pathlib import Path
from PIL import Image

image_path = Path(__file__).parent / "Assets" / "cat.jpg"
image = Image.open(image_path)

print(image.size)
print(image.filename)
print(image.format)

image.show()
```

```
leminh@Mac-2 Week_4_Lab_Activities % python3 intro.py
(1920, 2879)
/Users/leminh/Desktop/Week_4_Lab_Activities/Assets/cat.jpg
JPEG
leminh@Mac-2 Week_4_Lab_Activities %
```

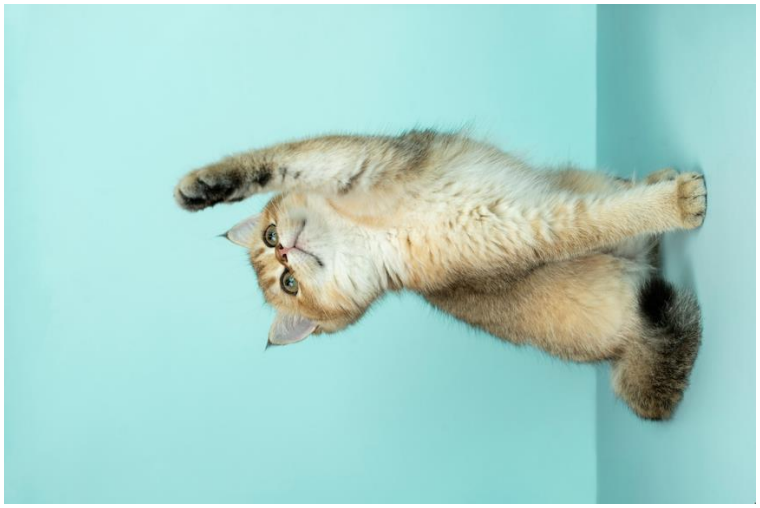
- Flipping the Image

```
-    from pathlib import Path
-    from PIL import Image
-
-    image_path = Path(__file__).parent / "Assets" / "cat.jpg"
-    image = Image.open(image_path)
-
-    # Print some information about the image
-    print(image.size)
-    print(image.filename)
-    print(image.format)
-
-    image.show()
-
-    # Rotate the cat image by 90 degrees
-    flipped_horizontal = image.transpose(Image.Transpose.FLIP_LEFT_RIGHT)
-    flipped_horizontal.show()
-    flipped_vertical = image.transpose(Image.Transpose.FLIP_TOP_BOTTOM)
-    flipped_vertical.show()
```

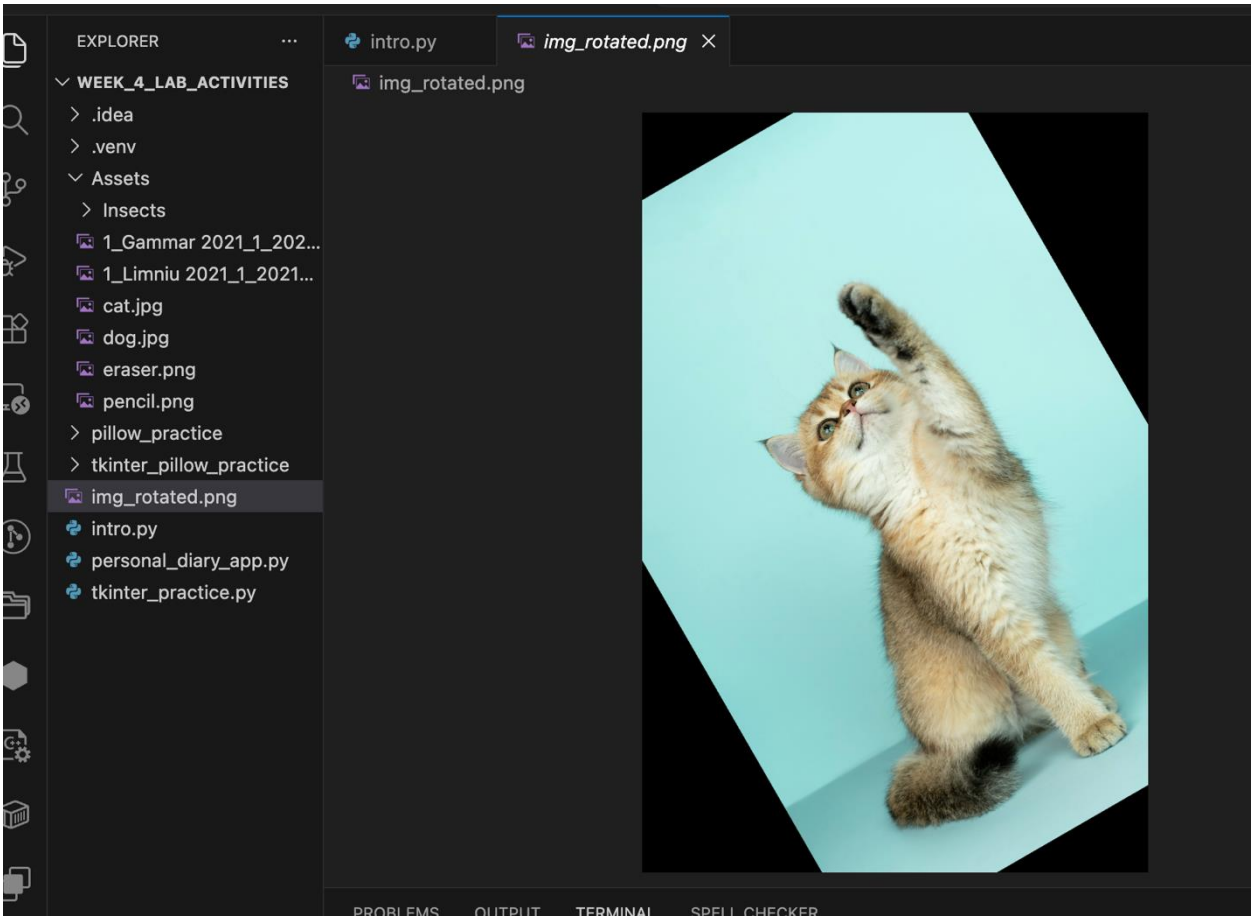


- Rotating the Image 90 degree

```
-    # Rotate the cat image by 90 degrees
-    rotated_image = image.transpose(Image.Transpose.ROTATE_90)
-    rotated_image.show()
```



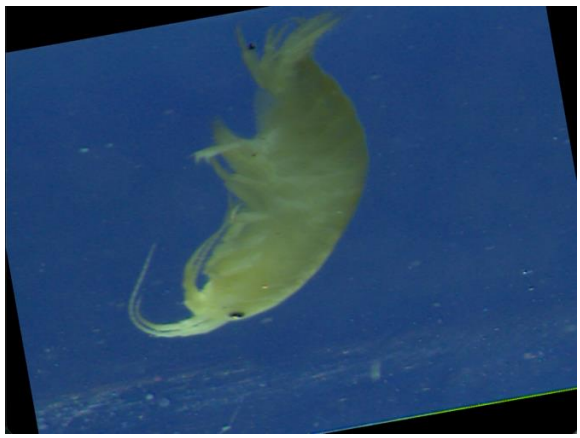
- Exercise: Load and Rotate a new Image and Save the Result



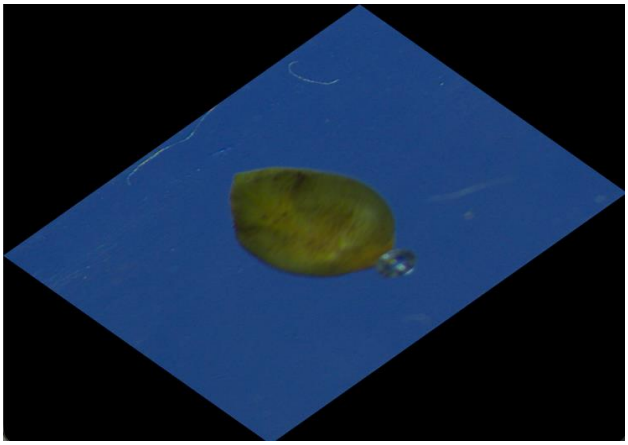
Task 3: Manipulating Images with Pillow/PIL library functions

Explanation: Apply image transformations such as rotate, crop, resize, and combine multiple operations to modify images programmatically.

```
from PIL import Image

# import an image
image = Image.open('Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.PNG')
# flip
image_transpose = image.transpose(Image.Transpose.ROTATE_90)
# rotation
image_rotate = image.rotate(45, expand = False, center = (0,0))
# crop
image_crop = image.crop((800,600,1600,1600))
# resize
image_resize = image.resize((1000,600))
# applying multiple transformations to an image in a single line of
#code
combined_image =image.transpose(Image.Transpose.ROTATE_90).resize((2000,1500)).rotate(10)
combined_image.show()
```



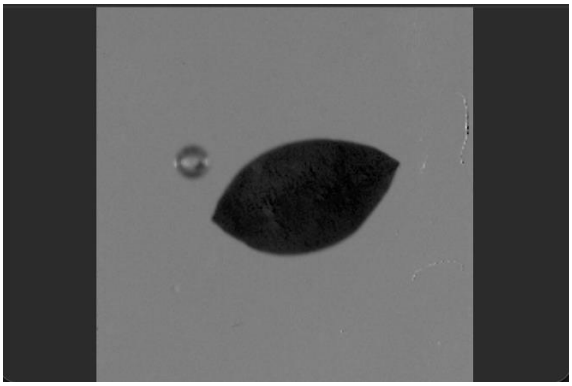
Case 2:

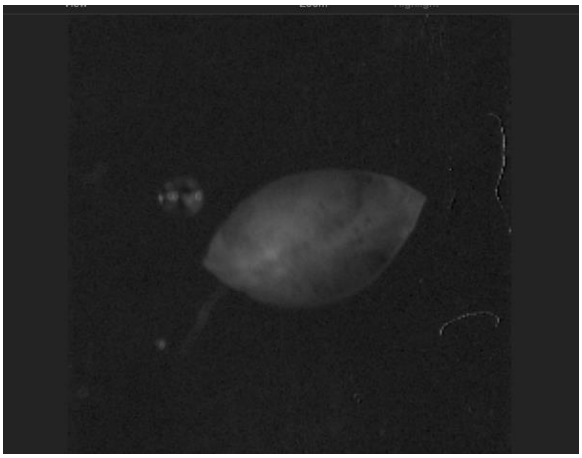


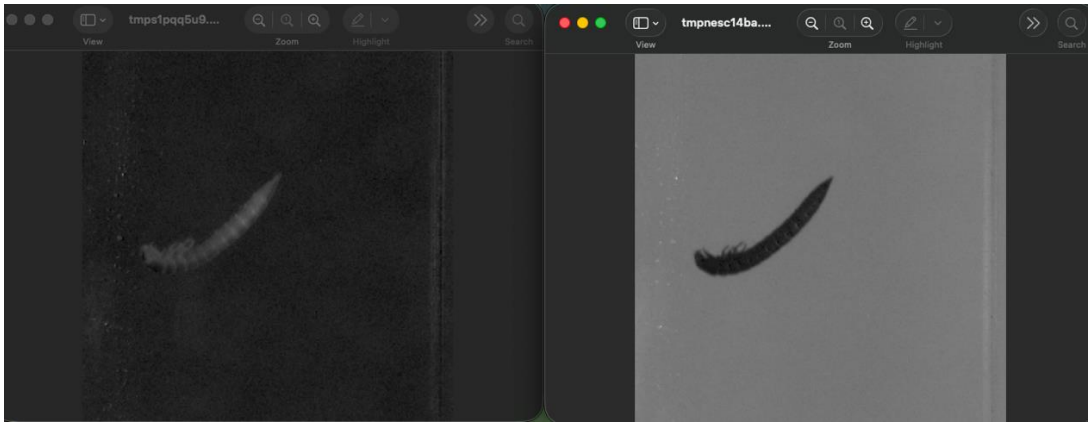
Case 3:

Task 4: Understanding Pixels and Colors with Pillow/PIL library functions

Explanation: I explored how images are made of pixels and RGB values, and learn to read and modify pixel colors using Pillow functions.

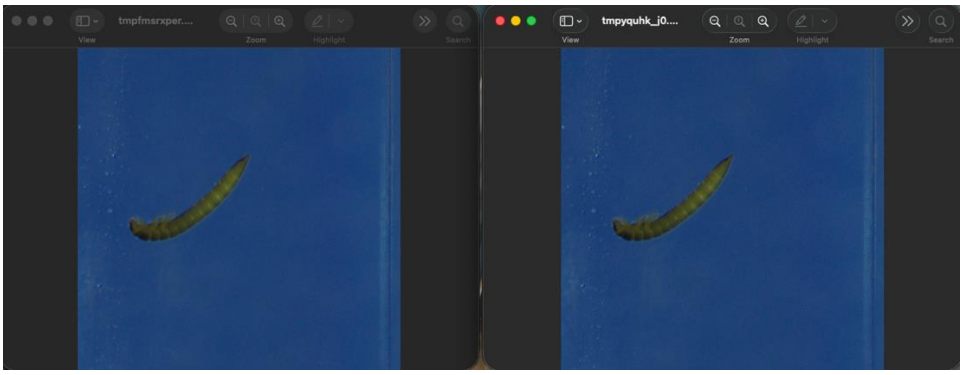


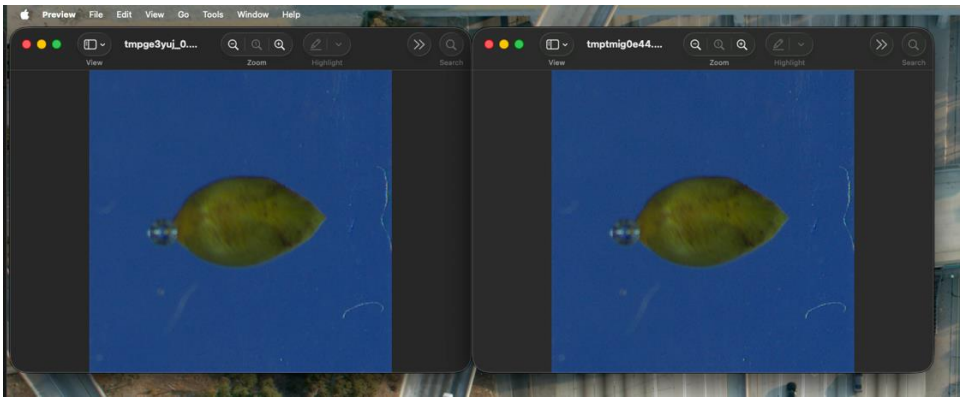


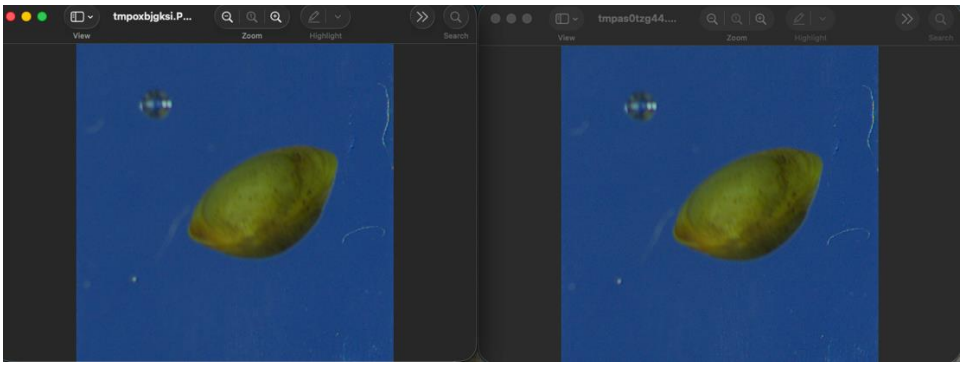


Task 5: Image Enhancement with Pillow/PIL library functions

Expanation: Use ImageEnhance to adjust properties like brightness, contrast, color, and sharpness to improve image quality.

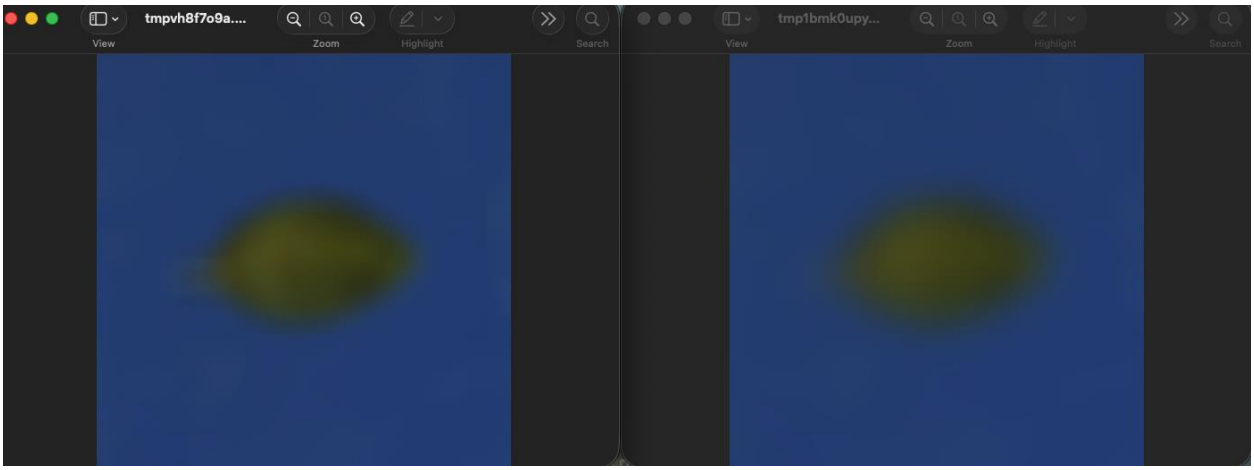


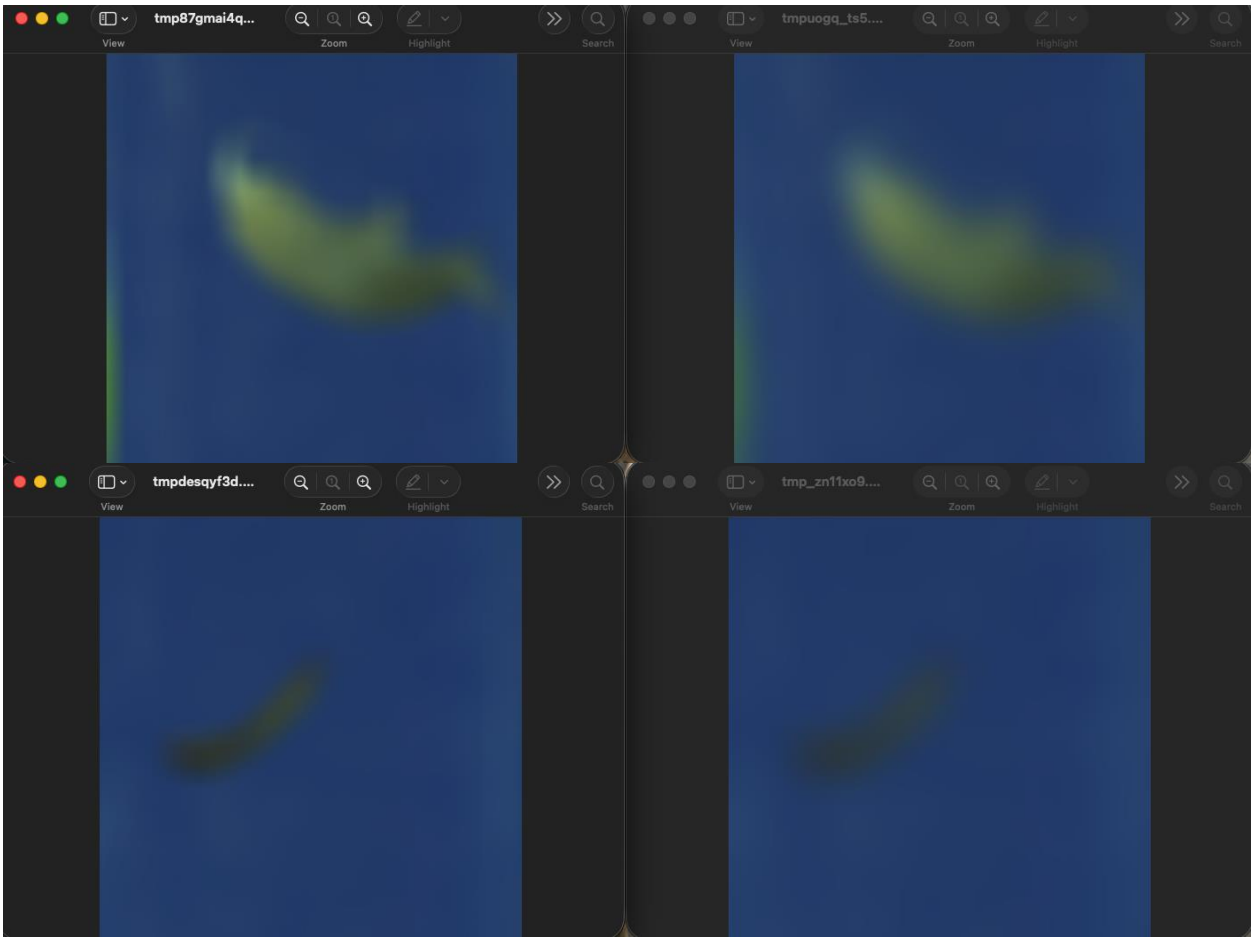




Task 6: Image filtering and Edge detection with Pillow/PIL library functions

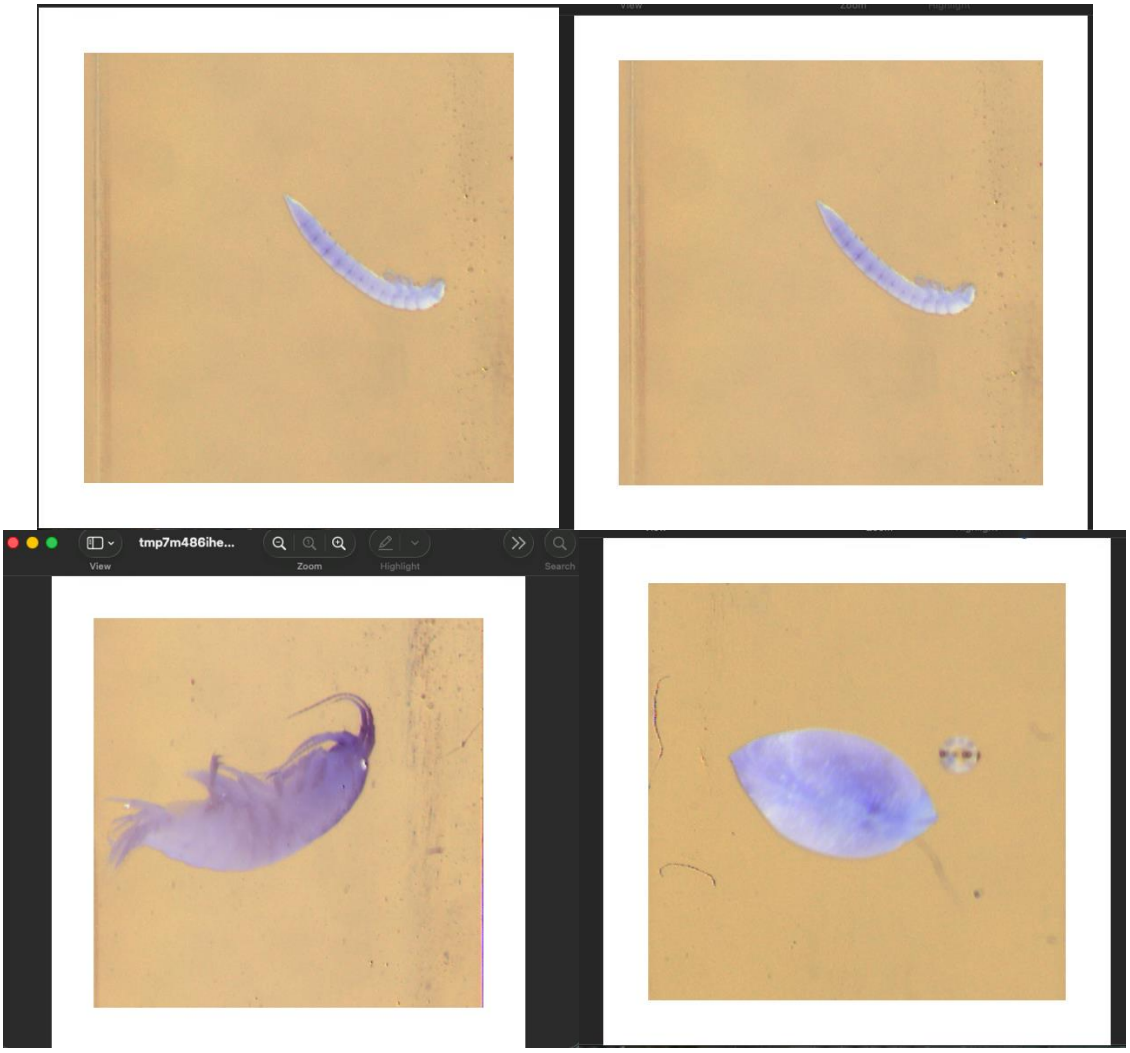
Explanation: This one taught us how to apply filters such as blur, contour, emboss, and edge detection to highlight or transform image features.





Task 7: Image Operations with Pillow/PIL library functions

Explanation: In this task I use ImageOps to perform operations like mirroring, inverting colors, and adding borders to images.


tmp7m486ihe...
View
Zoom
Highlight
Search